

PROFESSIONAL EXPERIENCE

- **Meta, Inc.** Menlo Park, US
Senior Machine Learning Engineer @ Instagram-Threads Feed Relevance Mar. 2026 – Present
 - I support new and low signal users for Threads growth, optimizing across the stack (retrieval, ranking model, value model, re-rank, cold-start).
 - I improve user experience in the re-rank layer as the technical lead, driving interest matching and diversity through end to end optimization.
- **TikTok, Inc.** San Jose, US
Senior Machine Learning Engineer @ TikTok Data-Plus Jan. 2025 – Mar.2026
 - I was the technical lead of TikTok's content push recommendation, I supported end to end optimization including **ANN-based retrieval, cold-start embedding modeling, multi-stage ranking, and LLM-based creative optimization.**
 - I made **20+ algorithm launches** at US scale, improving personalization quality and driving **+0.4% DAU** (~1M users) incremental **user growth.**
 - I mentored **10+ junior engineers and interns** across Shanghai and US teams. I helped to scaled the team from 2 to 20.
- **ByteDance (TikTok)** Shanghai, CN
Senior Machine Learning Engineer @ TikTok Data May. 2022 – Jan. 2025
 - I was the technical lead of the social recommendation modeling and founding member of the group. I scaled TikTok's **friends recommendation system (people you may know)** from 0 to 1;
 - I made 30+ launches per year across **offline/online retrieval, end-to-end ranking, and rec sys strategies.** I contributed 80%+ of our group goal per year. Achieved **300% growth in business metrics** and contributed **~1%+ DAU lift**, collaborating cross-functionally with backend, infra, and product teams.
 - I proposed and implemented a **real-world-social network project**, introducing relationship feedback loops and hybrid retrieval mechanisms. Led a collaboration team with over **20 people**, resulting in **15+ impactful launches** and **0.5% dau lift.**
- **Tencent, Inc.** Shenzhen, CN
Machine Learning Researcher @ IEG Lightspeed&Quantum Studios (LOL Mobile) Apr. 2021 – May. 2022
 - Developed **new-player skill detection system** for League of Legends Mobile; iterated 10+ model versions across client/server, reducing one-sided matches by **20%** and improving retention by **+1% (30M DAU).**
 - Built a **transformer-based matchmaking model with multi-task learning**, predicting game quality and fairness; increased player satisfaction by **10%**, reduced one-sided games by **20%**, and improved retention by **1.5%**
 - Delivered **data infra tools** (Spark/Go/K8s dashboards, user profiling services), resolving 50+ tickets and supporting multiple game features. Awarded **company-level business breakthrough award.**

EDUCATION

- **University of California, San Diego** San Diego, US
Master of Science in Electrical and Computer Engineering Sept. 2018 – Dec. 2020
- **Northwestern Polytechnical University** Xi'an, CN
Bachelor of Science in Aerospace Engineering Sept. 2014 – Jun. 2018

PUBLICATIONS

- **Kai Wang**, Ge Fan, Chaoyun Zhang, Yuyang Jiang, and Yuze Liu. CHAMP: Cross-domain Hybrid Architecture for Matchmaking and Prediction in Online Multi-Player Games. *CIKM*, 2026.
- **Kai Wang**, Chaoyun Zhang, Hao Chen, Ge Fan, Yingjie Li, Lifang Wu, and Bingchao Zheng. Quickskill: Novice skill estimation in online multiplayer games. *CIKM*, 2022.
- **Kai Wang**, Chunxu Shen, Chaoyun Zhang, and Wenye Ma. AdnFM: An attentive densenet based factorization machine for CTR prediction. *ICCDE*, 2022. **Best Oral Presentation Award.** [Paper]

- Ge Fan, Chaoyun Zhang, **Kai Wang**, and Junyang Chen. Mv-han: A hybrid attentive networks based multi-view learning model for large-scale content recommendation. *ASE*, 2022.
- Ge Fan, Chaoyun Zhang, and **Kai Wang**. Cupid: Improving battle fairness and position satisfaction in online MOBA games with a re-matchmaking system. *CSCW*, 2024.

PROFESSIONAL SERVICES

- **PC** CIKM 2024, CIKM 2025, CIKM 2026